My Computer Vision & Multimodal Work + Bridging Agentic AI, 3D Reconstruction & Adaptive Learning — A Technical Overview

Before diving in, I want to take a moment to sincerely apologize for not clearly communicating my computer vision background earlier in our conversation.
I came into the interview very "wired" around agentic systems, LangGraph graphs, and real-time voice simulation because I had been immersed in that work intensely, and I didn't surface the CV work I actually spent a significant part of my time on.

To make up for that, I've put together a structured overview of my experience in 3D reconstruction, neural rendering, multimodal generation, and psychometric modelling, along with how these directly map to Convergent's ambitions in adaptive learning, simulation realism, and embodied AI.

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1. COMPUTER VISION WORK (PRESENZ)

This was my most research-heavy CV phase, deeply focused on scene reconstruction, neural rendering, and reflective-surface modelling — all of which map directly onto simulation technologies.

4D Gaussian Splatting (Dynamic Scene Reconstruction)

I worked extensively with spatiotemporal Gaussian Splatting (GS), combining spatial splats with temporal features to reconstruct dynamic scenes from videos.

In practice, this meant:

- Breaking video into frame sequences
- Approximating each frame as a multi-view input
- Aligning temporal splats for stable motion representation
- Producing a 4D representation that captured motion, lighting, and geometry consistently

This allowed dynamic subjects — people, objects, motion — to be reconstructed photorealistically and in real time, enabling synthetic training data generation

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Multi-View Stereo (MVS) Reconstruction

I designed and refined pipelines around COLMAP-style MVS, Depth-from-Focus, and deep-learning-based multi-view stereo networks.

In this phase, I built:

- High-resolution 3D meshes
- Stable geometry under varied motion
- Reconstructions robust to occlusion, lighting changes, and reflections

My contribution involved:

- Tuning depth-fusion parameters and surface thresholds
- Minimising reprojection error
- Ensuring consistency across sparse and dense view samples

I also worked deeply with the intermediate geometric layers:

- Managing and optimizing PLY files
- Performing GS → mesh conversions without losing detail
- Implementing depth-map fusion to reduce noise
- Refining surface normals and topology

A significant part of the pipeline was pose estimation.
I experimented with:

- Classical bundle adjustment
- Learned pose regressors
- Hybrid optimization techniques

This experience gave me a strong intuition for how fragile 3D reconstruction pipelines can be, and how to design them to be robust, predictable, and production-ready.

Neural Radiance Fields (NeRF)

I implemented NeRF pipelines with:

- Positional encoding
- Differentiable volumetric rendering
- Hierarchical sampling
- Camera-pose optimisation
- Radiance-field regularisation

This work enabled the generation of high-fidelity, novel views of scenes and objects, which directly translated into stronger simulation and training capabilities. By reconstructing environments with photorealistic detail, I could support virtual training spaces, enrich scenes through augmentation, create immersive simulation experiences, and produce synthetic datasets that improved model robustness.

This helped me develop a deep understanding of:

- Implicit neural representations
- Density and radiance modelling
- Differentiable rendering

All are directly relevant to embodied AI simulations.

Reflective Object Rendering

My most challenging task was building a pipeline for rendering reflective, metallic, and glass surfaces — one of the hardest CV problems.

What I built:

- Used pretrained Gaussian Splat models
- Manipulated PLY-based representations
- Built a dynamic view-angle rendering controller
- Optimized spherical harmonics (SH) coefficients for accurate reflections
- Produced photorealistic renderings without needing to retrain the full GS model

Applications included:

- Quality inspection
- Product visualization
- Synthetic training assets

This work required an understanding of:

- Non-Lambertian surfaces
- View-dependent radiance
- Neural rendering theory
- Real-time rendering optimisation

This area is especially relevant to Convergent because high-fidelity, realistic environments dramatically improve training immersion and behavioural simulation outcomes.

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2. COREWARE — Big Five Personality Modelling + Multimodal Generation

At Coreware, I worked on controllable personality-driven generation, exploring how AI can adapt behaviorally to human traits.

Big Five Trait Modelling

I built systems for:

- Single-Trait Shaping
  Modulating output based on one psychological dimension.
- Multi-Trait Shaping
  Controlling expression across the five major personality factors:
  Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism.

This required an understanding of psychometric scoring, embeddings, human behavior modelling, and interpretability.

Multimodal Model Work

Models I used:

- HuggingFace's DALL-E
- Stable Diffusion Models and viT models
- CLIP (vision-language alignment)
- VQ-VAE-2
- BigGAN
- GPT-based text generators

I implemented:

- Trait-controlled text & image generation
- Latent-space conditioning
- Prompt modulation based on personality weights
- Consistency scoring under psychometric constraints

This experience maps directly to Convergent's behavioural scoring, coaching, and adaptive feedback systems.

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3. HOW THIS WORK APPLIES TO CONVERGENT

3D Reconstruction for Simulation Environments

Convergent could incorporate:

- Scene reconstruction
- Spatially accurate training spaces
- Photorealistic environment generation
- Embodied agent interactions
- Spatial memory for LLM agents

GS and NeRF pipelines could radically enhance:

- Training realism
- Scene diversity
- Multimodal feedback
- Trainee immersion

Video-Based Adaptive Coaching

My CV background enables Convergent to incorporate:

- Gesture recognition
- Gaze tracking
- Facial micro-expression analysis
- Posture and movement evaluation
- Audio-visual fusion models

This would allow Convergent to score not just *what* a trainee says, but *how they behave* — dramatically improving skill assessment, like the example you gave earlier today – an AI agent can show different facial expressions – Anger, Worry, Confused, etc.

Embodied AI + Agentic Reasoning

With my LLM-agent + 3D reconstruction experience, Convergent can build:

- Spatially grounded interactive agents
- Context-aware coaching inside virtual spaces
- Agents that react to user body language, gaze, and motion
- Next-generation simulation pipelines

New SOTA Trends (2024–2025) Relevant to Convergent

- 4D Gaussian Splatting
- GS 2.0 + fast SH optimisation

- NeRF-based avatar generation
- MVSNet, Depth Anything V2
- ViT-based trackers
- Video-grounded LLM reasoning
- Multimodal LMMs (Qwen2-VL, Gemini Pro 2)
- OpenAI Realtime + vision fusion
- Embodied AI frameworks (Habitat 3.0, HoloAssist)
- Neural rendering for real-time simulation

I’ve worked directly with several of the underlying technologies (GS, NeRF, MVS, SH), that would help me extend them to Convergent’s ecosystem.

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4. COMMITMENT TO LEARNING CONVERGENT'S STACK

Even though I haven’t yet seen the full internal stack, I can confidently say:

- I learn new frameworks extremely quickly
- I have worked across LLMs, CV, multimodal, and agentic pipelines
- I am excited to contribute to Convergent’s next evolution
- I thrive in high-speed, ambiguous, problem-solving environments
- I’m deeply motivated to help push adaptive learning into its next frontier

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CLOSING NOTE

Thank you again for the conversation — it genuinely made me even more excited about Convergent’s vision. I hope the current submission was up to the standards, and again, I am more than willing to make any further changes you might have.
I hope this document clarifies the depth of my computer vision and multimodal experience, and how I can bring that into Convergent’s adaptive, immersive, simulation-driven future.

I’m happy to elaborate on any technical details.